COURSE TRACKER: SHERYIANS AI SCHOOL

Complete Statistics for Data Science — Study Guide

Full Course Summary: Descriptive & Inferential Stats, Z-Scores, Probability, Bayes' Theorem, Hypothesis Testing & Decision Rules.

1. Descriptive vs. Inferential Statistics

Descriptive Statistics

What it is: Describing what is already in your data.

Simple Terms: Like taking a snapshot of a class's test scores to find the average grade and score range.

Key Tools: Mean, Median, Mode, Variance, Standard Deviation, Histograms, Box Plots.

Inferential Statistics

What it is: Making smart predictions about a whole group using a small sample.

Simple Terms: Tasting one spoonful of soup to decide if the entire pot needs more salt.

Key Tools: Hypothesis Testing, Confidence Intervals, p-values, Z-tests, t-tests.

2. Measures of Central Tendency & Spread

- **Mean:** The arithmetic average (Sum / Count). Sensitive to extreme outliers.
- **Median:** The exact middle value when data is sorted. Robust against outliers.
- **Mode:** The most frequently occurring value in the dataset.
- **Variance & Standard Deviation (σ):** Measure how spread out the data points are from the mean. Standard Deviation is simply the square root of Variance.
- **Interquartile Range (IQR):** The distance between the 25th percentile (Q1) and 75th percentile (Q3). Used in Box Plots to detect outliers (**Q1 − 1.5 × IQR** or **Q3 + 1.5 × IQR**).

3. Z-Scores & Standardization

A **Z-score** tells you how many standard deviations a data point is away from the mean.

$$Z = (X - \mu) / \sigma$$

In Simple Terms: If your test score has a Z-score of +2.0, you scored 2 standard deviations higher than the class average!

4. Probability & The Multiplication Rule

Measures the likelihood of events occurring together: **P(A ∩ B)**.

Independent Events

One outcome does *not* affect the other (e.g., rolling a die and flipping a coin).

$$P(A \cap B) = P(A) \times P(B)$$

Dependent Events

The first outcome changes the probability of the second (e.g., picking cards without replacement).

$$P(A \cap B) = P(A) \times P(B|A)$$

Conditional Probability P(B|A): "Chance of B happening, *given that* A has already happened." Formula: **P(B|A) = P(A ∩ B) / P(A)**.

5. Bayes' Theorem

A formula to update your beliefs when new evidence comes in.

```
P(A|B) = [ P(B|A) × P(A) ] / P(B)
```

> **In Simple Terms:** "What is the chance I have a disease (A) given that my test came back positive (B)?" You update your prior risk with test accuracy.

- **P(A) [Prior]:** Initial belief before new evidence.
- **P(B|A) [Likelihood]:** Chance of seeing this evidence if belief A is true.
- **P(A|B) [Posterior]:** New updated belief after seeing the evidence.

6. Hypothesis Testing & Decision Rules

A formal statistical method to test if an idea or effect is real or just random luck.

- **Null Hypothesis (H_0):** The boring status quo ("Nothing changed", "No effect", "Dog did not steal cookies").
- **Alternative Hypothesis (H_a):** The claim you want to prove ("There is a real effect", "Dog stole cookies").
- **Significance Level (α):** Your risk limit for making a mistake (usually set to 0.05 or 5%).

Type I vs. Type II Errors

Decision Made	H_0 is ACTUALLY TRUE	H_0 is ACTUALLY FALSE
Reject H_0	**Type I Error (α)** False Alarm (Accusing innocent dog)	**Correct Decision** True Positive
Fail to Reject H_0	**Correct Decision** True Negative	**Type II Error (β)** Missed Detection (Guilty dog gets away)

7. p-Value vs. Critical Value Methods (Opposite Approaches)

Both methods lead to the exact same conclusion, but they look at the decision from opposite sides:

1. P-Value Approach (Look for SMALLER)

Compare: p-value vs. Alpha (α)

Rule: If **p-value ≤ α**, **REJECT H_0**.

Why: Small p-value means the evidence is too rare to be a coincidence.

Code Note: `1.197e-05` means `0.00001197` (Scientific Notation). Since 0.00001197 ≤ 0.05, you REJECT H_0!

2. Critical Value Approach (Look for BIGGER)

Compare: Calculated Z/T score vs. Critical Value

Rule: If **Test Score > Critical Value**, **REJECT H_0**.

Why: A huge score means your sample result traveled deep into the "rejection zone".

> **Golden Takeaway:** P-Value method wins by going **UNDER** the limit (≤ α). Critical Value method wins by going **OVER** the limit (> Critical Value).